

Conventions: $\epsilon_{\mu_1 \dots \mu_n} = \sqrt{-g} \epsilon_{\mu_1 \dots \mu_n}$

• n-forms:

$$X \equiv x^{\mu_1 \dots \mu_p} \sqrt{-g} (d^{n-p} x)_{\mu_1 \dots \mu_p}$$

$$(d^{n-p} x)_{\mu_1 \dots \mu_p} = \frac{1}{p!(n-p)!} \epsilon_{\mu_1 \dots \mu_p \nu_{p+1} \dots \nu_n} dx^{\nu_{p+1}} \wedge \dots \wedge dx^{\nu_n}$$

• A vector field j^μ will be regarded as an $(n-1)$ form

$$J \equiv j^\mu \sqrt{-g} (d^{n-1} x)_\mu$$

$$\delta x \phi^i = + \mathcal{L}_X \phi^i$$

Surface Charges in Gravitation:

• Special Rel.:

The symmetries of the spacetime are defined as vector fields ξ^μ that preserve the background structure:

$$\mathcal{L}_\xi g_{\mu\nu} = 0$$

$$\begin{aligned} \mathcal{L}_\xi g_{\mu\nu} &= \xi^\alpha \partial_\alpha g_{\mu\nu} + g_{\mu\alpha} \xi^\alpha \partial_\nu + g_{\alpha\nu} \partial_\mu \xi^\alpha \\ &= \xi^\alpha \partial_\alpha g_{\mu\nu} + 2 \partial_{[\mu} \xi^{\alpha} g_{\nu]\alpha} \end{aligned}$$

→ Note, these are Killing vector fields

$\Leftrightarrow$

$$g_{\mu\nu}(x') = g_{\mu\nu}$$

$\exists$ an isometry

$$g_{\mu\nu}(x) = \frac{\partial x'^\alpha}{\partial x^\mu} \frac{\partial x'^\beta}{\partial x^\nu} g(x')$$

$$x'^\mu = x^\mu + \xi^\mu$$

$$g_{\mu\nu}(x) = \frac{\partial (x^\alpha + \xi^\alpha)}{\partial x^\mu} \frac{\partial (x^\beta + \xi^\beta)}{\partial x^\nu} g(x + \xi)$$

* Since this is a flat spacetime, we find $\partial_\alpha \eta_{\mu\nu} = 0$ and are left with

$$\begin{aligned} \partial_\mu \xi^\alpha \eta_{\mu\alpha} + \partial_\nu \xi^\alpha \eta_{\mu\alpha} &= 0 \\ \partial_\mu \xi^\alpha \eta_{\mu\alpha} + \partial_\nu \xi^\alpha \eta_{\mu\alpha} &= 0 \\ \partial_\mu \xi^\alpha \eta_{\mu\alpha} + \partial_\nu \xi^\alpha \eta_{\mu\alpha} &= 0 \end{aligned}$$

$$\begin{aligned} \partial_\mu X^\kappa g_{\kappa\nu} + \partial_\nu X^\lambda g_{\mu\lambda} \\ \partial_\mu (X_\sigma \eta^{\kappa\sigma}) \eta_{\mu\alpha} + \partial_\nu (X_\sigma \eta^{\lambda\sigma}) g_{\mu\lambda} &= 0 \\ \partial_\mu X_\sigma \delta^\sigma_\nu + \partial_\nu X_\sigma \delta^\sigma_\mu &= 0 \\ \boxed{\partial_\mu X_\nu + \partial_\nu X_\mu = 0} \end{aligned}$$

* Constant Solution:

- $\xi^\mu_{(i)} = \delta^\mu_{(i)} \quad (0 \leq i \leq 3)$ [Spacetime Translations]
- Let $\xi^\mu = b^{\mu\nu} X_\nu \Leftrightarrow \xi_\mu = b_{\mu\nu} X^\nu$
 $\partial_\mu (b_{\nu\alpha} X^\alpha) + \partial_\nu (b_{\mu\alpha} X^\alpha) = 0$
 $b_{\nu\alpha} \partial_\mu X^\alpha + b_{\mu\alpha} \partial_\nu X^\alpha = 0$
 $b_{\nu\alpha} \delta^\alpha_\mu + b_{\mu\alpha} \delta^\alpha_\nu = 0$
 $b_{\nu\mu} + b_{\mu\nu} = 0$
 $\Rightarrow b_{\mu\nu} = -b_{\nu\mu}$
 $b_{\mu\nu} = b[\mu\nu]$

Isometries: $\xi^\mu = a^\mu + b^{\mu\nu} X_\nu$
 $b_{\mu\nu}$ has $\binom{4}{2} = 6$ terms

3: $\xi^\mu_{(i)} = 0, \xi^\mu_{(i)} = \delta^\mu_{(i)} X^\nu \quad (1 \leq i, \mu, \nu \leq 3)$

Rotations about the x^i axis

3: $\xi_{(a)0} = x^a$ $\xi_{(a)m} = -\delta_{am} x^0$ ($1 \leq a, m \leq 3$)
Lorentz Boosts along x^a axis.

* Under a Lie Bracket these give rise to the Poincaré Alg.

* In an arbitrary coordinate system we have
 $\partial_\mu \rightarrow \nabla_\mu$

Killing Equation:

A necessary and sufficient condition for a vector field to be an isometry of a spacetime (M, g) is that it verifies the Killing Eq. $\mathcal{L}_\xi g_{\mu\nu} = 0 \iff \nabla_{(\mu} \xi_{\nu)} = 0$

Existence of a conserved stress tensor:

Any rel. field theory in M^4 admits a symm. stress tensor $T^{\mu\nu}$ that is divergence free when eq^s of motion hold:

$$\nabla_\mu T^{\mu\nu} \approx 0$$

* $d^n x \mapsto \sqrt{-g} d^n x$

$$T^{\mu\nu} := \frac{2}{\sqrt{-g}} \frac{\delta L}{\delta g_{\mu\nu}}$$

$$\delta L / \delta \phi^i := \frac{\delta L}{\delta \phi^i} - \partial_\mu \left(\frac{\delta L}{\delta \partial_\mu \phi^i} \right) + \partial_\mu \partial_\nu \left(\frac{\delta L}{\delta \partial_\mu \partial_\nu \phi^i} \right) + \dots$$

(Euler-Lagrange derivative)

• Most see $T^{\mu\nu}$ is covariantly conserved

$$\rightarrow \delta L = \delta \phi^i \frac{\delta L}{\delta \phi^i} + \partial_\mu \Theta^\mu \left(\frac{\delta L}{\delta \partial_\mu \phi^i} \right)$$

$$= \delta \phi^i \frac{\delta L}{\delta \phi^i} + \partial_\alpha \Theta^\alpha [\delta \phi, \phi]$$

where Θ^α is all of the boundary terms

• Divergence of a vector field density

$\Theta^\alpha [\delta \phi^i, \phi^i]$ called the bare

presymmetrized potential

or

$$\delta \mathbb{L} = \delta \phi^i \frac{\delta \mathbb{L}}{\delta \phi^i} + d \Theta [\delta \phi, \phi]$$

where

$$\Theta = \Theta^\mu (d^{n-1} x)_\mu, \quad \mathbb{L} = L d^n x$$

where

$$d \Theta = dx^\mu \partial_\mu [\Theta^\nu (d^{n-1} x)_\nu] = dx^\mu \partial_\mu \Theta^\nu (d^{n-1} x)_\nu$$

$$= \partial_\mu \Theta^\mu d^n x$$

* We can look at the variation of $\mathbb{L}$ under infinitesimal diffeos generated by ξ^μ

$$\delta_\xi \mathbb{L} = \delta_\xi g_{\mu\nu} \frac{\delta \mathbb{L}}{\delta g_{\mu\nu}} + \delta_\xi \phi^i_\mu \frac{\delta \mathbb{L}}{\delta \phi^i_\mu} + d(\dots)$$

$$= \mathcal{L}_\xi g_{\mu\nu} \frac{\delta \mathbb{L}}{\delta g_{\mu\nu}} + \delta_\xi \phi^i_\mu \frac{\delta \mathbb{L}}{\delta \phi^i_\mu} + d(\dots)$$

$$= 2 \nabla_\mu \xi^\mu \frac{\delta \mathbb{L}}{\delta g_{\mu\nu}} + \delta_\xi \phi^i_\mu \frac{\delta \mathbb{L}}{\delta \phi^i_\mu} + d(\dots)$$

Recall $T^{\mu\nu} = \frac{2}{\sqrt{-g}} \frac{\delta \mathbb{L}}{\delta g_{\mu\nu}}$, thus

$$\delta \mathbb{L} = \sqrt{-g} T^{\mu\nu} \nabla_\mu \xi_\nu d^n x + \delta_\xi \phi^i_\mu \frac{\delta \mathbb{L}}{\delta \phi^i_\mu} + d(\dots)$$

* For M^4 : if we assume we are on shell:
 Thus $\frac{\delta \mathcal{L}}{\delta \phi^i} = \frac{\partial \mathcal{L}}{\partial \phi^i} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi^i)} \right) + (\dots)$

Second term vanishes

$$\begin{aligned} \delta \mathcal{L} &= \sqrt{-g} T^{\mu\nu} \nabla_\mu \xi_\nu d^n x + d(\dots) \\ &= \nabla_\mu (\sqrt{-g} T^{\mu\nu} \xi_\nu d^n x) - \nabla_\mu (\sqrt{-g} T^{\mu\nu}) \xi_\nu d^n x + d(\dots) \\ &\approx \sqrt{-g} d^n x \nabla_\mu T^{\mu\nu} \xi_\nu - \sqrt{-g} d^n x \nabla_\mu T^{\mu\nu} \xi_\nu + d(\dots) \\ &\approx \sqrt{-g} d^n x \left(\partial_\mu T^{\mu\nu} \xi_\nu + \Gamma_{\mu\beta}^\mu T^{\beta\nu} \xi_\nu \right) - \sqrt{-g} \dots \\ \text{Assuming that } \Gamma_{\mu\beta}^\mu &= 0 \text{ ???} \\ &\approx d^n x \partial_\mu (\sqrt{-g} T^{\mu\nu} \xi_\nu) - \sqrt{-g} d^n x \nabla_\mu T^{\mu\nu} \xi_\nu + d(\dots) \end{aligned}$$

* LOCALLY we can assume that the first term can be written as an exact form. i.e. $\leftarrow$ on-shell. Thus

$$\delta \mathcal{L} \approx \sqrt{-g} d^n x \nabla_\mu T^{\mu\nu} \xi_\nu + d(\dots)$$

• The theory/Covariance requires that the total variation of $\mathcal{L}$ must be a total derivative $\delta \mathcal{L} = d(\dots)$

In which the contents of the $(\dots)$ must have compact support within the boundary ∂M .

$$\delta \mathcal{L} = d(\dots) \Leftrightarrow \nabla_\mu T^{\mu\nu} \Big|_{g_{\mu\nu} = \eta_{\mu\nu}} \approx 0$$

* We can now build the current density from $T^{\mu\nu}$:

its Hodge dual:

$$\vec{J} = j^\mu \text{fg}(d^{n-1}x)_\mu$$

Provided that the diffeo g^μ is an isometry of spacetime:

$$\nabla_\mu j^\mu = \nabla_\mu T^{\mu\nu} g_\nu + T^{\mu\nu} \nabla_\mu g_\nu = 0 + 0$$

$$\Rightarrow d\vec{J} = 0$$

* $\vec{J}$ on an arbitrary Cauchy surface Σ produces a scalar quantity

$$Q := \int_\Sigma \vec{J}$$

which is conserved when the system evolves in time.

- $x^0 = t$, choose Σ s.t. the surface at $t = 0$: $(d^{n-1}x)_0 = d\Sigma$

$$\partial_t Q = \partial_t \int_\Sigma \vec{J}^0 (d^{n-1}x)_0 = \int_\Sigma \partial_0 j^0 d\Sigma$$

But

$$\nabla_\mu j^\mu = 0 = \partial_0 j^0 + \vec{\nabla}_i \cdot \vec{J} = 0$$

thus

$$\partial_t Q = - \int_\Sigma d\Sigma \vec{\nabla} \cdot \vec{J} = - \int_\Sigma \vec{J} \cdot d\vec{S} = 0$$

Isometry	Origin	Conserved charge
Translations	M^4 is homogeneous	$P^\mu = \int_\Sigma d\Sigma T^{\mu 0}$
Lorentz trans.	M^4 is isotropic & relativistic	$M^{\mu\nu} = \int_\Sigma d\Sigma (x^\mu T^{\nu 0} - x^\nu T^{\mu 0})$

Generalized Noether's Theorem

Global Symmetries: Preserve $\mathbb{L}$ up to a boundary term.

Gauge trans: global symm whose generators arbitrarily depends upon the coords.

Noether's first thm

Take $\mathbb{L}$ defined on (M, g) that admits global symm (some of which may be gauge):

$\exists$ a bijection between

(a) The equivalence class of global continuous symmetries of $\mathbb{L}$, and

(b) The eq. classes of conserved vector fields J^μ , the so called Noether Currents

(a) Two global symm of $\mathbb{L}$ are equivalent iff they differ by a gauge trans. $\ni$ Another symm whose gen. is trivially zero on-shell.

$$(g - g')\phi = e^{\hat{T}}\phi$$

where $\hat{T}|_{E.O.M.} = 0$

(b) Two currents j_1^μ , j_2^μ are eq. if they differ by a trivial current of the form

$$j_2^\mu - j_1^\mu = \partial_\nu K^{[\mu\nu]} + t^\mu$$

$$K^{\mu\nu} \in \Delta^2(M) \quad \text{and} \quad t^\mu \approx 0 ???$$

$\left\{ \begin{array}{l} \text{is that to say } |t| \ll 1 ??? \\ \text{or is that to say } t^\mu \text{ is a constant} \end{array} \right.$

$$\text{Thus: } \partial_\mu (\partial^\mu \phi - \partial^\mu \phi) = \partial_\mu \partial_\nu K^{[\mu\nu]} + \partial_\mu t^\mu = 0 \Rightarrow \partial_\mu \partial^\mu \phi \approx \partial_\mu \partial^\mu \phi$$

* We now encounter a problem:

(i) Suppose we have a pure gauge theory

• From Noether's theorem, $\exists$ exists only one eq. class of conserved currents:

The trivial ones.

• For general covariant theories, $x^\mu \mapsto x^\mu + \xi^\mu$ is a gauge-transformation.

and so isometries are associated to trivial currents in a similar way

$$Q = \int_\Sigma \partial^\mu (\partial^\mu \phi) = \int_\Sigma (\partial_\nu K^{[\mu\nu]} + t^\mu) (d^{n-1}x)_\mu$$

($t^\mu \approx 0$????)

$$Q \approx \int_\Sigma \partial_\nu K^{[\mu\nu]} (d^{n-1}x)_\mu \approx \int_{\partial\Sigma} K^{[\mu\nu]} (d^{n-2}x)_{\mu\nu}$$

Thus, Q is completely arbitrary since $K^{\mu\nu}$ is unconstrained

* Explicitly: $\mathcal{H} = \left(\frac{R\sqrt{g}}{16\pi G} + L_m \right) d^4x$

$$T^{\mu\nu} = \frac{2}{\sqrt{g}} \frac{\delta \mathcal{H}}{\delta g_{\mu\nu}} = \frac{\delta(R\sqrt{g})}{\delta g_{\mu\nu}} \left(\frac{1}{16\pi G} \right) + T_m^{\mu\nu}$$

$$= - \frac{G_{\mu\nu}}{8\pi G} + T_m^{\mu\nu}$$

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R$$

∴

$$T^{\mu\nu} = - \frac{1}{8\pi G} (R^{\mu\nu} - \frac{1}{2} g^{\mu\nu} R) \approx 0$$

Thus, there is a trivial conserved current

$$j^\mu = T^{\mu\nu} \xi_\nu \approx 0$$

Lower Degrees Conservation Laws: